



Characterizing and Stabilizing Drive Shaft

MECHENG 495: Lab II

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BACKGROUND

The Gulo 3000 exhibits transient vibrations during DC motor start-up and shut-down. Preliminary investigation suggests that flexibility in the drive shaft contributes to the observed vibration.

APPROACH

- Characterize the transient response of the motor/drive shaft system
- Develop and validate a dynamic model of the experimental system
- Use the model to identify design modifications capable of reducing vibration by $\geq 30\%$

CONCLUSIONS & RECOMMENDATIONS

- Drive-shaft flexibility contributes significantly to the transient vibrations observed during motor start-up and shut-down
- The developed dynamic model successfully represents the behavior of the experimental motor/drive system
- The validated model can be used to evaluate design modifications without requiring prototypes for each configuration
- Several design modifications were identified that achieve the required $\geq 30\%$ reduction in vibration amplitude.

Recommended Design Modifications

- Idea 1
- Idea 2
- Idea 3
- Idea 4

ACKNOWLEDGEMENTS

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RESULTS

A

Image of damping (maybe the zoom in, it'd be good to show our first period i think.)

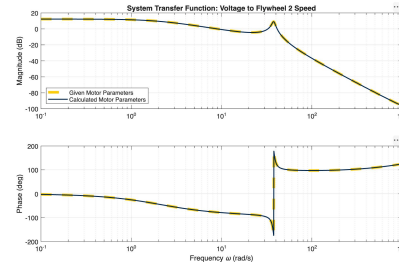
B

Image of sweep (i think it'd be good to show the harmonic frequency)

Fig 2. Transient Plotting. The experimentally determined transfer function closely agrees with the expected second-order system response. This agreement validates the developed dynamic model and indicates that the model adequately captures the dominant dynamics of the motor/drive-shaft system.

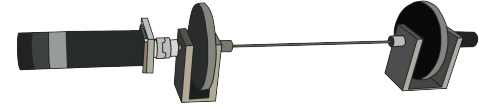
Fig 1. Experimental Characterization. (A) An eraser was used to constrain the wheel and induce an oscillatory response in the drive shaft. Peak-to-peak measurements were used to determine the damping ratio, and frequency ratio, providing the damping characteristics required for the dynamic model

(B) A frequency sweep identified the system's harmonic frequency, providing the frequency response necessary for model characterization



FUTURE STUDIES

- Evaluate additional shaft materials and their effects on stiffness and dynamic response
- Investigate the effect of shaft diameter and geometry on vibration amplitude
- Another design change
- One more change to the system



Labels will be added once we know what to address

Fig 3.

I ASSUME we will be changing the shaft diameter, material, length, or the wheel diameters. This is the part where 30% fixing will go

Fix 1

Fix 2

Fix 3

Fix 4

REFERENCES

- [1] MathWorks, MATLAB, The MathWorks, Inc., Natick, MA.
- [2] Servo-TEK. 1086 Goffle Road, Hawthorne, NJ 07506. 2003. PDF